

Analysis Report: Multimodal Fake News Detection Baseline and Evaluation of an Existing Deepfake Detection System

In Support of the TruthLens AR Project

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Abstract—This report presents two studies informing TruthLens AR's design. First, a multimodal fake news classifier combining BERT and ResNet50 via late fusion on Fakeddit reaches 77.46% accuracy. Second, an existing open-source EfficientNet-B0 deepfake detector is shown to repeatedly misclassify modern AI-generated images as authentic with high confidence. Gaps identified, missing OCR, fact-checking, reverse image search, Bangla support, and explainability, motivate TruthLens AR's architecture.

Index Terms—*Fake News Detection, Deepfake Detection, Multimodal Learning, BERT, ResNet50, Late Fusion, EfficientNet, Vision-Language Models, Bangla NLP*

I. INTRODUCTION

As groundwork for TruthLens AR, a Bangla multimodal misinformation detection system, two exercises were undertaken: benchmarking a Fakeddit-trained multimodal classifier to evaluate fusion strategies, and evaluating an existing open-source deepfake detector to identify failure cases and feature gaps. Together they establish an evidence base for TruthLens AR's design.

II. BASELINE MULTIMODAL FAKE NEWS CLASSIFICATION

A. Problem Motivation and Dataset

Traditional detection datasets rely almost exclusively on text, yet roughly 87% of misinformation is carried through images or video, often pairing misleading text with manipulated images. This motivates using Fakeddit, a multimodal benchmark of over one million text-image samples with six fine-grained class labels and full metadata.

B. Model Architecture: Late Fusion of BERT and ResNet50

The model processes text and images through two independent encoders. A BERT encoder produces 768-dimensional semantic embeddings capturing fake linguistic patterns, while a ResNet50 encoder produces 2048-dimensional embeddings sensitive to image artifacts and manipulation cues.

The modalities are combined via late fusion: each encoder produces a 6-class prediction, merged by element-wise maximum before final softmax classification, letting the stronger modality's confidence dominate.

C. Fine-Grained Classification Scheme

The model distinguishes six fine-grained categories of news content, summarized in Table I.

Class	Description
TRUE	Authentic, fact-checked content
SATIRE	Intentional parody or comedy
FALSE CONNECTION	Misleading headline, unrelated image
IMPOSTER	Impersonates legitimate sources
MANIPULATED	Edited/manipulated media
MISLEADING	Distorted, sensationalized content

TABLE I. SIX FINE-GRAINED FAKE NEWS CLASSES

D. Performance Results

The multimodal model reached 77.46% accuracy, outperforming both unimodal baselines by 6–9 points (Table II), confirming text-image inconsistencies carry information neither modality captures alone.

Approach	Accuracy
Text-only (BERT)	68.2%
Image-only (ResNet50)	71.5%
Multimodal (Late Fusion)	77.46%

TABLE II. UNIMODAL VS. MULTIMODAL ACCURACY

Per-class accuracy: Satire 92%, Manipulated 85%, False Connection 74%, Misleading 69% (hardest, needing subtler semantic reasoning).

E. Prototype Web Application

A Streamlit prototype accepts a headline and image and returns a 6-class prediction. Qualitative testing on Bangla and English pairs showed correct classification of some samples but misclassified others, indicating room for improvement in fusion and Bangla robustness.

III. ANALYSIS OF AN EXISTING DEEFAKE DETECTION SYSTEM

A. System Overview

An existing open-source deepfake detector built on pretrained EfficientNet-B0 was evaluated to identify gaps TruthLens AR should address. It runs end-to-end but initial testing suggested it struggles with modern AI-generated images.

B. Practical Run Results

Two known-fake AI-generated images were submitted: the system predicted "real" with 87.06% and 99.86% confidence respectively, indicating a consistent high-confidence false-positive pattern.

C. Identified Limitations

- No fine-tuning for modern diffusion-generated imagery, causing weak, overconfident results
- No OCR, reverse image search, or fact-check API integration
- No Bangla support or explainable AI (XAI) output
- Limited robustness and generalization overall

IV. DISCUSSION: IMPLICATIONS FOR TRUTHLENS AR

The two studies show complementary gaps: late-fusion struggles on subtler classes, favoring a joint Vision-Language Model, while the existing EfficientNet-B0 deepfake detector, unlike the proposed system, is image-only and fails confidently on AI imagery with no OCR, fact-checking, reverse search, explainability, or Bangla support. Together they justify TruthLens AR's VLM reasoning and full verification suite.

V. CONCLUSION

This report combined a Fakeddit baseline and a deepfake-detector evaluation to build an evidence base for TruthLens AR, confirming simple fusion's value and limits and current detectors' gaps, supporting TruthLens AR's VLM, OCR, fact-checking, reverse search, explainability, and Bangla support.

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